

Tel: 608-886-6081 • E-mail: changhanc@berkeley.edu • Web: chang-han-chen.github.io

Education

- **Ph.D. in Theoretical Physics, University of California, Berkeley** expected **end of 2026**
- **B.Sc. in Physics and Mathematics, Massachusetts Institute of Technology** 2022
GPA **4.9/5**; Morse/Orloff award for best senior thesis.

Skills (besides physics)

- **Research:** von Neumann algebras, random matrix and free probability, language model training.
- **Coursework:** [algorithm design](#), [convex optimization](#), probability theory, statistical learning theory.

Research Interest

What is the most puzzling, mysterious aspect of our universe?

Universality of holography: $GR + QM = \text{hologram}$? Instead of betting on one particular theory such as string theory or loop quantum gravity that are “invisible” to current technology, I believe in the universality of holography: *any* non-perturbative quantum theory consistent with general relativity is holographic, i.e. the information in quantum gravity lives on the boundary. The goal of my Ph.D. is to explore as many aspects of this universality as possible, e.g. two-dimensional toy models and von Neumann algebras.

For my recent interest in AI, see [this blog](#).

Selected Projects

- **MoE autoresearch** 2026
Explored LLM-agent-driven “autoresearch” on autoregressive mixture-of-experts transformers, stacking interventions to a 1.75% bits-per-byte gain that persists under compute-optimal scaling.
- **Optimal block-size curriculum for diffusion language models** 2026 Spring
Designed a block-size curriculum for pretraining block-diffusion language models, sweeping the autoregressive-vs-diffusion training budget and finding a consistent sample-efficiency optimum near 30%.
- **An apologia for islands** (with S. Antonini, H. Maxfield, G. Penington) 2025 Spring
Proved that entanglement islands and the Page curve emerge even with *massless* gravitons and no external reservoir, using gauge-invariant gravitational operators to show black-hole interior operators are non-perturbatively encoded in Hawking radiation.
- **A clock is just a way to tell the time** (with G. Penington) 2024 Spring
Resolved prevalent misconceptions and pioneered practical applications of abstract mathematical tools to cosmological modeling, with work that became fairly influential in the field.
- **Geometry and entanglement in AdS/CFT and beyond** 2022 Spring
Independent senior thesis, awarded MIT’s Morse/Orloff prize for best senior thesis in theoretical physics.

Honors and Awards

- **Morse/Orloff Research Award** – best senior thesis in theoretical physics, MIT 2022
- **Gold medal, International Physics Olympiad (IPhO)** 2018

References

Prof. Hong Liu	Prof. Geoff Penington	Prof. Aron Wall
MIT	UC Berkeley	U. of Cambridge
hong_liu@mit.edu	geoffp@berkeley.edu	aronwall@gmail.com